

Alex Sugiarto

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PROFESSIONAL SUMMARY

Mechanical & Manufacturing Engineering graduate with hands-on experience in product design, CAD, and engineering documentation for manufacturing. Demonstrated experience delivering end-to-end product development from concept through to production, including technical drawings, BOMs, and production planning. Experience in additive manufacturing, CNC processes, and cost-controlled production, with proven ability to translate requirements into manufacturable solutions. Background includes process optimisation, Python coding and cross-functional collaboration with customers, suppliers, and production teams.

CAREER OBJECTIVE

Mechanical & Manufacturing Engineering graduate with hands-on experience in product design, manufacturing, prototyping, and process improvement. Seeking a graduate engineering role where I can apply my practical engineering skills, safety-focused mindset, and problem-solving abilities within a fast-paced manufacturing environment. Eager to contribute to continuous improvement initiatives, operational efficiency, and collaborative engineering projects while continuing to develop my technical and leadership capabilities.

EDUCATION

Bachelor of Engineering (Mechanical and Manufacturing) UNSW Sydney	2021 - 2025
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WORK EXPERIENCE

Prototyping Engineer KeenzKustoms	April 2025 - Current
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- Led small-scale engineering design and fabrication projects from concept to production with iterative design
- Reviewed client specifications and translated requirements into manufacturable solutions with CAD, 3D Printing and CNC
- Tracked project budgets and material costs to ensure financial control

Independent 3D Printed Products & Climbing Accessories E-commerce store	April 2025 - Current
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- Designed, prototyped, and manufactured custom 3D printed products using CAD and additive manufacturing techniques
- Generated \$75,000+ in revenue in first year as a solo founder through end-to-end product lifecycle management

- Managed concept development, prototyping, testing, production planning, and customer delivery

Manufacturing Technician

Jan 2019 - Feb 2020

Thai Surat Steel (TSS)

- Interpreted engineering drawings to ensure production accuracy
- Operated machinery in compliance with safety and quality standards
- Monitored production data using Excel and supported quality inspections

Project Internship

Dec 2024 - Feb 2025

Gate gourmet, Mascot Sydney

- Assisted in process optimisation and quality control initiatives with Arena software
- Supported documentation and workflow improvements

RELEVANT PROJECTS

Sculpture by the Sea - Assisted in design, fabrication and structural assembly of large-scale public installations

The Tuckerbox - Designed and fabricated an injection-moulded insulated picnic basket

Gearbox Design & Analysis -Designed gears and shafts, validated with ANSYS FEA

Crack detection system - Designed crack detection imaging software with Python numpy

I have many projects under my belt, portfolio available upon request

ADDITIONAL EXPERIENCE

Courier

Feb 2020 – Dec 2022

Uber / Menulog, Sydney

Waiter/Kitchen Hand

Feb 2023 – May 2024

Pinocchio, Sydney

Store vendor

Feb 2021 – April 2025

Classic Squeeze, Sydney

TECHNICAL SKILLS

ANSYS | Rockwell Arena | CAD | CNC | Python | Microsoft Excel, Word, PowerPoint | Engineering design | Additive manufacturing | Cost control | Quality support | Process improvement | Injection moulding

REFEREES

Contacts available on request